Supplementary File 1

The patient samples were donated by an asymptomatic 30-year-old woman carrying a high mutational load for the m.11778 locus. After ovarian stimulation, multiple cumulus oocyte complexes were retrieved, comprising metaphase II (MII) oocytes, one metaphase I (MI) oocyte, and one germinal vesicle (GV) oocyte. The GV oocyte was incubated for 24 h, and the MI oocytes was incubated for 24 h in in vitro maturation (IVM) medium under mineral oil at 37 °C in 6% CO2 and 5% O2. These conditions are hereafter called standard culture conditions. The IVM medium was prepared in-house by supplementing tissue culture medium-199 (TCM-199) containing Earle's salt and bicarbonate with 10 ng/ml epidermal growth factor (EGF), 1 µg/ml estradiol, 10 mIU/ml follicle stimulating hormone (FSH), 0.5 IU/ml hCG, 100 IU/ml penicillin G potassium salt, 100 µg/ml streptomycin sulphate salt, 0.3 mM sodium pyruvate and 0.8% human serum albumin. Four MII oocytes were subjected to fertilization by standard intracytoplasmatic sperm injection (ICSI) (samples 10-13). Five in vivo MII oocytes obtained from the patient were used for ST as nuclear donor (samples 1-5), while the MI oocyte was subjected to ePNT after in vitro maturation (IVM) (sample 6). ST was performed according to Tang et al. (2019), ePNT was performed according to Hyslop et al. (2016). Enucleated, frozen-thawed, in vivo matured MII oocytes with smooth endoplasmic reticulum aggregates (SERa) (Irvine Scientific, USA) donated by healthy donors served as cytoplast recipients for ST, while a healthy, enucleated, fresh IVM oocyte served as recipient for ePNT. Following NT, reconstituted embryos and control embryos were cultured in Cook Cleavage (day 1-3) or Cook Blastocyst (day 3-6) medium under mineral oil at standard culture conditions. ST resulted in samples 17-22, ePNT resulted in sample 23. Other assessed samples comprised a non-patient enucleated MII oocyte (sample 14), two patient zygotes (samples 8 and 9), and two non-patient IVM oocytes (samples 15 and 16).

Sources:

Tang M, Guggilla RR, Gansemans Y, Van der Jeught M, Boel A, Popovic M, et al. Comparative analysis of different nuclear transfer techniques to prevent the transmission of mitochondrial DNA variants. Mol Hum Reprod 2019;25:797-810.

Hyslop LA, Blakeley P, Craven L, Richardson J, Fogarty NM, Fragouli E, et al. Towards clinical application of pronuclear transfer to prevent mitochondrial DNA disease. Nature 2016;534:383-6.

Supplemental Table S1: Detailed overview of the analyzed samples.

Sample code	Sample description
1	Patient enucleated in vivo Metaphase II oocyte
2	Patient enucleated in vivo Metaphase II oocyte
3	Patient enucleated in vivo Metaphase II oocyte
4	Patient enucleated in vivo Metaphase II oocyte
5	Patient enucleated in vivo Metaphase II oocyte
6	Patient In-vitro maturation (MI to MII) enucleated oocyte
7	Patient germinal vesicle (GV) oocyte
8	Patient zygote (osteopontin)
9	Patient zygote (osteopontin)
10	Patient arrested embryo after in-vitro fertilization (Day 6)
11	Patient arrested embryo after in-vitro fertilization (Day 6)
12	Patient arrested embryo after in-vitro fertilization (Day 6)
13	Patient arrested embryo after in-vitro fertilization (Day 6)
14	Non-patient enucleated Metaphase II oocyte
15	Non-patient in vitro maturation (GV to MII) oocyte
16	Non-patient in vitro maturation (GV to MII) oocyte
17	Trophectoderm cells from a spindle transfer (ST) generated blastocyst (Day 5)
18	Trophectoderm cells from a spindle transfer (ST) generated blastocyst (Day 5)
19	Arrested embryo (16 cells) after spindle transfer (Day 5)
20	Arrested embryo (8 cells) after spindle transfer (Day 5)
21	Arrested embryo (5 cells) after spindle transfer (Day 5)
22	Arrested embryo (8 cells) after spindle transfer (Day 5)
23	Arrested embryo after early pronuclear transfer